

Tony Nguyen

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Education

B.A. in Computer Science & Computational Thinking, Data Science & Economics *Expected May 2026*
B.B.A in Business Administration, Finance

Gonzaga University, Spokane, WA
President's Honor List, GPA: 3.97

Research Interests

Knowledge representation and evaluation in LLMs. Retrieval-augmented generation, agentic systems, and semantic alignment metrics. Resource-constrained deployment and knowledge updating methods for reliable AI systems.

Research & Industry Experience

AI / Machine Learning Research Assistant (Advisor: Dr. Jay Yang) January 2025 - Present
Institute for Informatics & Applied Technology - Gonzaga University *Spokane, WA*

- Benchmarked **three LLMs** on **more than 60 CVE reports** using a **three-stage RAG pipeline** with specialized **agents** for retrieval, analysis, and remediation recommendation, focusing on **insightfulness and verifiability**.
- Investigated **semantic embedding** strategies to handle conflicting or out-of-distribution security documentation beyond base model knowledge, with emphasis on failure modes and uncertainty.
- Designed and deployed **MCP agent services** on university servers, exposing secure endpoints for external users with authentication and observability to support reliable, multi-tenant access and testing.
- Developed and delivered a workshop on **RAG systems** and **LLM evaluation** to over **30 participants**, covering classical metrics, agent-based evaluation, and responsible AI practices through live coding demos.

Data Engineer Intern June 2025 - September 2025
KPIM Vietnam *Hanoi, Vietnam*

- Designed and maintained **10+ data pipelines** using **SQL, Python, and Microsoft Fabric** for **2 Big Data projects**, enforcing complex schema validation, data quality checks, and optimizations to make querying **50% faster** than our client's internal system.
- Built reusable tooling and documentation to streamline troubleshooting, standardize workflows, and support stakeholders with reliable, self-serve financial and operational data of the client's **6 hotel properties**.

Data Engineer - Innovation Intern June 2024 - August 2024
KONE Elevators & Escalators *Singapore*

- Built a **centralized Customer 360** by integrating **20 years** of data from **5 countries**; aligned models to enterprise architecture and recommended schema optimizations to improve consistency and join performance.
- Developed a **Tableau** dashboard that fed from **SAP, Salesforce, and Excel**; created data models and calculations to monitor customer behavior, service KPIs, and revenue in real time.
- Engineered and operated **ETL pipelines** on **AWS S3** for ingestion, storage, and validation; automated integrity checks and partitioning to **reduce data retrieval latency by 40%** and improve reliability.

Data Science & Machine Learning Intern June 2023 - August 2023
TecAlliance Vietnam *Ho Chi Minh City, Vietnam*

- Engineered Python algorithms to parse data from **100+ Wiring Diagram SVGs**, creating structured datasets for computer vision models (**YOLO**) and advanced ML pipelines.
- Deployed a **Neo4j graph database** to store complex relationships extracted from automotive schematics, increasing **query speed by 80%** compared to the client's current database and providing deeper insights.
- Led weekly **global team meetings** of nearly **15 people** to unify data engineering efforts, consolidate findings, and manage development timelines across multiple time zones.

Conference Poster Presentations

- Value & Responsibility in AI Technologies Conference, 2025. Spokane, WA.
- Spokane Intercollegiate Research Conference, 2025. Spokane, WA.

Project

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- Minecraft AI Companion – Applied AI Developer, Senior Capstone Project** August 2025 - Present
- Built an open source Minecraft mod connected to locally hosted **LLMs via Ollama**, enabling world-aware villager dialogue; designed the LLM API, **chunk-aware context pipeline**, and **guardrails** for safe in-game interactions.
- Mortality Risk Simulator – Project Lead** January 2025 - May 2025
- Led development of a **neural-network-based mortality risk prediction pipeline** over demographic, vitals, and medical history features to support more accurate and fair underwriting review.

Teaching and Mentoring Experience

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- Teaching Assistant** August 2024 - Present
Department of Computer Science - Gonzaga University *Spokane, WA*
- Supported **150+** undergraduates across **3 data science courses**, leading weekly office hours to reinforce students' Python, pandas, and scikit-learn skills and core algorithms and data workflows.
 - Graded **10 programming assignments each term** using shared rubrics, providing targeted feedback on correctness, code structure, and documentation to improve code quality and conceptual understanding.
- Academic Tutor & Mentor** October 2022 - August 2024
Gonzaga Global - Gonzaga University *Spokane, WA*
- Provided tutoring and lab support for **7 computer science and quantitative courses** for **dozens of students each term**, contributing to **15%+ average grade improvement** for tutees.
 - Mentored international students on adapting to U.S. college coursework and expectations, including study strategies, communication with professors, and academic integrity.

Professional & Leadership Experience

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- Administrative Assistant** March 2022 - Present
Center for Global Engagement - Gonzaga University *Spokane, WA*
- Mentored **50+** incoming international students through cultural and academic transition to US college; designed automated workflows reducing administrative burden by **30%**, enabling more direct student support.
- Co-Founder & Senior Advisor** November 2021 - May 2025
Vietnamese Student Association - Gonzaga University *Spokane, WA*
- **Co-founded** the organization to foster supportive community for Vietnamese students; **mentored 10+ leaders** in data-informed planning, increasing event turnout by **20%** and generating **\$700+ profit**.

Professional Affiliations & Memberships

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- **Member**, *Society of Hispanic Professional Engineers*

Technical Proficiencies

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| Programming Languages | Python, SQL (MySQL, BigQuery SQL), C++, Java. |
| Data & ML Frameworks | pandas, NumPy, scikit-learn, PyTorch, spaCy, Streamlit, Flask, FastAPI, pytest. |
| Data Engineering & Analytics | Tableau, Power BI, Microsoft Fabric, Salesforce CRM, ETL pipelines, Docker, Git, Linux, Google Cloud, AWS. |
| AI & LLM Systems | RAG pipelines, LangChain, CrewAI, Semantic Kernel, Hugging Face Transformers, MCP, AI agents, embeddings. |

References

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- **Dr. S. Jay Yang**, *Director*
Institute for Informatics & Applied Technology, Gonzaga University
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 - **Dr. Gina Sprint**, *Associate Professor & Data Science Program Director*
Department of Computer Science, Gonzaga University
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 - **Dr. Shawn Bowers**, *Professor*
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